

ARTICLE



Real-time co-creation and nowness service: lessons from tourism and hospitality

Dimitrios Buhalis and Yeyen Sinarta

Head of Department of Tourism and Hospitality, Faculty of Management, International Centre for Tourism and Hospitality Research, Bournemouth University, Poole, UK

ABSTRACT

Brands take advantage of technology, social media and constant connectivity to foster organic consumer engagement and interactions towards co-creating personalised customer service. Real-time service offers dynamic engagement with connected consumers. Brands in tourism and hospitality use technology to dynamically enhance consumer experience through co-creation. The integration of real-time consumer intelligence, dynamic big data mining, artificial intelligence, and contextualisation can transform service co-creation by mobilising resources in the ecosystem. Nowness service emerges by dynamically engaging consumers in experience cocreation in real time. It has five interconnected characteristics that revolutionise the tourism and hospitality, namely: real-time, co-creation, data-driven, consumer-centric and experience co-creation.

ARTICLE HISTORY

Received 17 December 2018
Revised 19 January 2019
Accepted 20 February 2019

KEYWORDS

Real time; nowness; smart; interactive; social media; marketing; strategy; instant gratification; tourism; hospitality

Introduction, state-of-the-art and objectives

Technology has been leading a range of transformations in the marketplace. These include the interaction of people and organisation; interaction of people and information; interaction of people and technology; technological implementation; interaction of organisation and information; and interaction of organisation and technology (Buhalis & Law, 2008; Yuan, Tseng, & Ho, 2019). Increasingly consumers utilise smart devices to navigate destinations, make bookings and to access and navigate through travel information (Doricic, Komsic, & Markovic, 2019). The emergence of social media and Information Communication Technology (ICT) have revolutionised the way brands and marketers reach out and engage with consumers (Buhalis & Foerste, 2015; Fotis, Buhalis, & Rossides, 2011). Brands have in-depth intelligence of their consumers and their needs (Sigala & Chalkiti, 2015). This enables them to engage in a dialogue, develop their products and enhance the speed and agility of their services. Search engine marketing (SEM) has transformed online marketing, enabling tourism organisations to list products and services in search engines (Paraskevas, Katsogridakis, Law, & Buhalis, 2011). However, most of the content on websites is usually static, with the exception of availability and pricing.

Social media facilitate the search for travel information online (Sigala, Christou, & Gretzel, 2012; Xiang & Gretzel, 2010). Social media also facilitate the direct interaction between consumers and brands in a more personal way (Buhalis & Law, 2008; Gretzel & Fesenmaier, 2009; Law, Buhalis, & Cobanoglu, 2014; Sigala et al., 2012) and to co-create customised experiences (Neuhofer, Buhalis, & Ladkin, 2015, 2013; Rihova, Buhalis, Gouthro, & Moital, 2018; Rihova, Buhalis, Moital, & Gouthro, 2015). Sun, Fong, Law, and Luk (2016) indicated that hotel guests use three major types of websites, namely those of tourism suppliers, such as official websites of airlines and hotels, travel intermediaries including OTAs, bedbanks and metasearch engines and finally review and social media sites such as TripAdvisor. Technology supports prospective tourists to make informed travel decisions, adding to their customer satisfaction and reducing their risk and fear (Wang, Wu, Luo, & Lu, 2017). Yu Leung, Sun, and Bai (2019) illustrate that various social media platforms also have major impacts on both businesses and consumers. Not only they help in business management, but also are capable of changing consumers' attitudes and behaviours. Many tourism organisations use social media, often through storytelling, to promote engagement, amusement and trigger emotional interaction (Tiago, Couto, Faria, & Borges-Tiago, 2018). Brands also utilise private

channels as a medium to communicate, engage and deliver messages, such as: WhatsApp, Facebook Messenger, Twitter Direct Message, or email newsletters. The impact of technology and the engagement in social media often depends on the location of service providers. In line with digital divide in tourism (Minghetti & Buhalis, 2010) it is evident that airlines from interned advanced countries tend to adopt technology and social media earlier than those in developing countries and perform better than their competitors (Baghirov, Zhang, & Hashim, 2019).

Social media can be used as a real-time interactive channel to co-create value with consumers and develop authentic engagement (Howell, 2014; Prahalad & Ramaswamy, 2004a; Sarmah, Kamboj, & Kandampully, 2018). Brands increasingly rely on online communities to engage in digital co-creation. This helps them to improve customer support, engage in meaningful relationships with customers, and develop innovative product and service development ideas (Zhang, Bilgihan, Kandampully, & Lu, 2018). A study by Evergage in 2014 showed that 76% of more than 250 marketers employ real-time marketing in their strategy and 48% of them utilise social platforms. Yet surprisingly, only 7% are able to provide data-driven and real-time engagement across both digital and face to face interactions (CMO Council, 2017).

ICTs, mobile technology and social media facilitate personalised and contextualised experience creation (Buhalis & Foerste, 2015). Consumer demand for instant gratification has been fuelled by the ability to interact in real time across different platforms. Highly informed consumers manage to co-create better value and are willing to spend more money with organisations that meet their requirements instantly (Darmer & Sundbo, 2008; Pine & Gilmore, 1999). Social media and mobile technology (mobile devices or mobile apps) empower productive dynamic interactions and instant gratification. Consumers are no longer only “consuming” the service, but increasingly participate in their own experience co-creation process. This provides them with valuable insights for brands to create and develop highly personalised services that meet their individual needs in real-time (Neuhofer et al., 2013; Sashi, 2012). Most brands have published widely accessible and visible social media channels and public usernames such as @Marriott or @Ryanair. Consumers therefore expect to be able to use it to communicate as they did with call centres in the past. This challenges brands to perform both proactively and reactively in a timely manner. Instant gratification means that price or

differentiation strategies can be irrelevant if organisations are unable to serve their consumers in real time (Christopher, Payne, & Ballantyne, 2002).

Technology has transformed value creation from a product-centric process to a consumer-centric approach

Increasingly “informed, networked, empowered consumers increasingly co-create value” using brands (Prahalad & Ramaswamy, 2004a, p. 5). Co-creation, defined as the “joint creation of value by the company and the customer” (Prahalad & Ramaswamy, 2004a, p. 8), reengineers the entire range of business processes. An experience ecosystem is emerging to facilitate value co-creation and to enable consumers to co-construct individualised, contextualised and personalised experiences (Buhalis & Foerste, 2015; Prahalad & Ramaswamy, 2004a). This interactive communication stream serves as a process for the co-creation of value and collaborative products or services (Prahalad & Ramaswamy, 2004b; Sawhney, Verona, & Prandelli, 2005) leading to more positive consumer behaviour towards the brand. Social media and mobile applications act as interactive platforms for brands to engage directly with consumers. This enables them to create dialogue, rectify any issues publicly and helps shaping the consumer journey dynamically (Court, Elzinga, Mulder, & Vetvik, 2009). Brands are faced with highly dynamic and unfiltered social conversations that influence consumer perceptions, attitudes and ultimately purchase intentions (Schniederjans, Atlas, & Starkey, 2018).

The travel, tourism and hospitality industries are experiential and dynamic in nature (Fotis, Buhalis, & Rossides, 2012; Hays, Page, & Buhalis, 2013). This brings expectations to focus service offerings to consumer needs, using contextualised and highly personalised service and engagement. Consumers’ personal preferences, combined with external factors including socio-political changes as well as weather, traffic or season, provide contextual information needed to co-create real-time value (Buhalis & Foerste, 2015; Prahalad & Ramaswamy, 2004b). Brands can differentiate themselves from competitors, in this highly competitive marketplace, by co-creating exceptional consumer experiences (McKinsey, 2016). This is particularly the case in New Product Development (NPD) when consumers can assume an active role in the offering firms ideas and solutions and helping in the development of new concepts often transforming business models and business offerings. Co-creation thus represents a particular form of open innovation (Costa & Coelho Do Vale, 2018).

This study aims to analyse how real-time big data and dynamic data-mining, based on contextual data, can facilitate instant value co-creation for stakeholders. It explores how social media and other technology platforms can enhance consumer experience in real-time. The study makes a contribution to marketing theory by elaborating on real team brand cocreation whilst provides management and marketing tools to practitioners to improve interactivity and competitiveness. It examines how the combination of technology, real-time big data and context develops opportunities for brands to co-create value for their digitally advanced consumers through nowness service (Buhalis & Foerste, 2015; Neuhofer, Buhalis, & Ladkin, 2012). The core objective of this paper is to explore “real-time service” and support the development of a real-time service ecosystem, through examining nowness and real-time interactions through social media and other digital platforms.

Defining real-time marketing

The concept of real-time marketing has been discussed for more than 20 years, supporting that flexible service offerings and dynamic timing can lead brands to enhance and increase their overall competitiveness and market place (Evergage, 2014; McKenna, 1995; Oliver, Roland, & Sajeev, 1998; Reid, 2014; Rust & Oliver, 2000). However, real-time marketing was only possible after the emergence of dynamic customer-provider interaction and social media, particularly Twitter and Facebook (Fotis et al., 2011). Veloutsou and Guzmán (2017) suggest “the imperative of brands adapting and adopting new technologies to keep up with market change is now that brands are technological to provide a seamless and real time brand experience throughout all real and virtual brand touchpoints.” Brands are expected to offer consumers with services or products that meet their present needs and adjust to changing demands or contexts (Urde, 2016; McKenna, 1997). Table 1 illustrates various definitions of real-time marketing. In essence, real-time marketing covers personalisation or mass customisation and relationship marketing and it is focused in the present through a range of contextualised variables that determine the co-creation environment. The timing and type of communication are also critical. Chandler, Salvador, and Kim (2018) show the effects of the corporate Twitter language types on firm value.

Real-time marketing requires continuous conversation with targeted consumers and dynamic engagement with the most influential audiences (Yoo & Gretzel, 2010) to meet consumer’s demand for instant gratification. It is a paradigm shift in service marketing

evolving from Service-Dominant (S-D) logic (Vargo & Lusch, 2004) embracing value-in-context as a value driver. Increasingly informed, connected, and knowledgeable consumers engage in conversation and assume the co-creator role in their value and experience co-creation (Prahalad & Ramaswamy, 2004a). Consumers collaborating with multiple actors enables them to optimise the value proposition and customise products or services to their personal needs and preferences (Chandler & Vargo, 2011; Schade, Piehler, Warwitz, & Burmann, 2018). Six dimensions of service appropriation, namely: service knowledge; service consciousness; self-adaptation to service; service control; service creation; and psychological ownership of the service can be applied to different contexts and objects (spaces, objects, feelings, experiences) (Mifsud, Cases, & N’Goala, 2015).

Consumers are increasingly at the centre of the brand strategy (Brogan & Smith, 2009; Christopher et al., 2002; Harwood & Garry, 2017). Dynamic communications enable data mining to create meaningful real-time contextual information and establish the conditions for nowness service. Brands can contextualise, personalise and individualise service offerings, using real-time big internal and external contextual data (Buhalis & Foerste, 2015). Co-creating value and experiences in real time require shifting the focus on channels and platforms where consumers are ready to engage instantly. Digital technologies, ambient intelligence, the Internet of Things (IoT) and the Internet of Everything as well as social media & mobile applications become the basis to fulfil the delivery of a brand’s promise, utilising real-time interactivity (Harwood & Garry, 2017; Porter & Heppelmann, 2014; Wilson, Zeithaml, Bitner, & Gremler, 2012; Tuškej & Podnar, 2018).

Therefore, real-time marketing can be defined as *marketing that provides personalised, individualised and contextualised products and services, based on real-time dynamic engagement with customers and co-creation of experiences, to optimise value for all stakeholders involved*. Real-time marketing is propelling “nowness” or present moment service and experience. This may complement or replace face to face experiences, such as speaking to a receptionist in a hotel, with digital services delivered online. These services can be offered by digital personnel such as online concierge services, applications and robots or friends, fellow travellers or experts. Dynamic big data mining and Artificial Intelligence (AI) as well as Emotional Intelligence (EI) and empathy are required to identify what really matters on that particular moment. Increasingly, bots and AI are being used to understand context, interpret signals and facilitate communication with all stakeholders.

Table 1. Definitions of real-time marketing.

Author/Institution	Definition
SmartFocus (Ca.2017)	"the ability to engage with your customers or fans instantly based on real-time information – like their actions and behaviour, changes to your own data or external news or events. It is instant, (mostly) automated, and targeted."
Econsultancy (2015)	"real-time marketing involves taking in real-time input from a customer and delivering a highly personalised, relevant and timely message back to them."
TranslateMedia (2015)	"Real-time marketing is a timely, agile style of promotion focused around current events and immediate feedback from customers."
Deutsch (2014)	"simply getting the message to the right person at the right instant. It is a way to interact with followers by focusing on something ongoing."
Lieb (2014)	"The strategy and practice of responding with immediacy to external events and triggers. It's arguably the most relevant form of marketing, achieved by listening to and/or anticipating consumer interests and needs."
Firebelly (2014)	"...what happens when a company uses real-time data to pivot their content, conversation, community and conversion approaches across social (owned, earned and paid) and other marketing channels to take advantage of time sensitive opportunities. It could be an effort towards being relevant or it could have a number of different objectives."
Smart Insights (2014)	"marketing that links into current events that are relevant to an audience in order to engage them with a brand."
Garner (2013, p. 3)	"a way of thinking and philosophy that requires businesses to meet the demands of an always-on digital world, and includes production, communication, organisation and infrastructure." "...is about time, existence in time, and using search and social technologies to interact and strike with lightning speed and <i>laser-like</i> efficiency."
Scott (2012)	"Real-time means news breaks over minutes, not days. It's when companies develop (or refine) products or services instantly, based on feedback from customers or events in the marketplace. And it's when businesses see an opportunity and are the first to act on it."
Rust and Oliver (2000, p. 55)	"Individually customised service product that tracks changing customer needs continuously and update itself to meet customer's personal needs through interaction with the customer and environment."
Oliver et al. (1998, p. 31–32)	"The marketing approach in which personally customised goods or services and continuously update themselves to continuously track changing customer needs, without intervention by corporate personnel, often without conscious or overt input from the customer."
McKenna (1995)	"replacing broadcast mentality that has long dominated marketing with a willingness to give consumers access to the company and to view their actions and feedback as integral to developing or improving products." "focusing on real-time customer satisfaction, providing support, help, guidance and information necessary to win customer's loyalty."

Equally, fellow travellers, local residents or travel experts can also contribute, often using hashtags, to identify a discussion topic and a point of interest. Present time is critical in real-time marketing and nowness, as this is what makes it relevant, contributes value and achieves instant gratification. The same information or value proposition may be totally irrelevant later to a particular traveller, although it may become as part of value proposition suite of solutions offered to people in similar situations.

Figure 1 shows six use-cases for best industry practice in real-time marketing. They fall into two axes, namely: proactive-reactive and planned unplanned (Lieb, Groopman, & Li, 2013). The highest control of real-time marketing occurs when efforts are categorised as planned/proactive quadrant. This includes *brand events* (e.g. product launches or conferences) and *anticipated events* (e.g. Starbucks prepared location-based and contextual-based promoted tweets to offer hot coffee when a snowstorm hits North America) (Heine, 2013). *Location-based* triggers serve highly relevant and personalised offers, combining location and physical factors (e.g. a restaurant or hotel) and real-time virtual data input (e.g. consumer "check-in" in a particular location via their mobile device). For example, Yelp partners with business owners to offer personalised deals based on user's locations and check-in action via an app, all in real-time (Yelp, 2010). The next one in the upper left quadrant, *Predictive analytics*, applies predictive data to provide

relevant recommendations to consumers based on their online journey. This can be combined with or complemented by consumer's location. Brands possess little control of content and conversations when the situations fall in reactive/unplanned sector of the real-time marketing quadrant. *Breaking news* requires a real-time response in a relevant manner to unanticipated news or events. This is arguably the most spontaneous and challenging for brands to implement as little advance planning can be prepared. The last one in the cluster, *customer interactions*, performs a critical role in real-time service, as consumers increasingly expect brands to respond to their online queries and complaints or praises instantly (Baer, 2013). Customer interactions, customer service, complaint handlings and community engagement are becoming the focus of marketing strategy due to the very public, highly visible and sporadically viral nature (Lieb et al., 2013). Therefore, the reputation of organisations and their competitiveness are assessed on their ability to co-create value in unplanned situations by understanding the deep motivations and desires of all actors in their ecosystem and dynamically come up with instant solutions that satisfy all.

Delivering the right content to the right consumer at the right time, especially on unplanned circumstances, requires a lot more than merely, observing, listening, designing, testing and creating expected service. It is caused by consumers' unique and dynamic personas, that are all influenced by their present context and

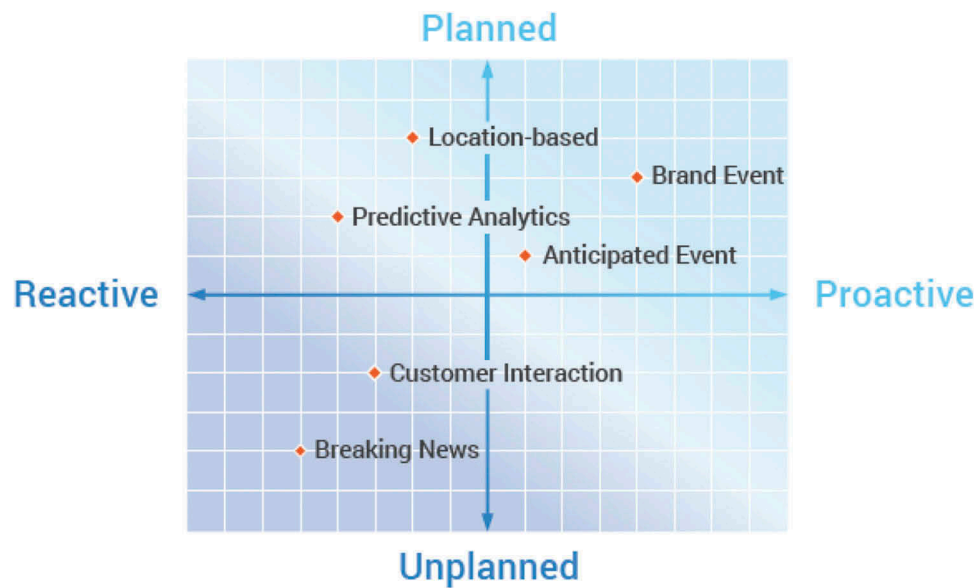


Figure 1. Six use cases of real-time marketing.

Source Lieb et al. (2013)

location. Brands need to understand the present activity or purpose of the activity before executing any action (Girish, 2016). Appreciating consumers' behavioural motivations, preferences and desires against all elements of their internal and external context is critical. The relevance and effectiveness of the nowness service needs to complement real-time data mining of consumers' online journey with contextual and situational factors that may affect their motivation, expectations and experience (Buhalis & Foerste, 2015). Proactively monitoring contextual situations in the planned journey, and dynamically analysing data, facilitates predicting unplanned events, risks and eventualities that influence customer experience. For example, arriving to a destination that is experiencing a major crisis may require the rerouting of travellers or the readjustment of itineraries even before the traveller is aware of the crisis. Monitoring weather condition between departure and arrival points may also result in the change of route or timing or even the cancellation of the trip. The dynamic arrangement of alternatives to optimise value through proactive, agile management is paramount for brands to offer highly personalised products and information and achieve competitiveness. Mobile Marketing Association (MMA), 2014.

Real-time co-creation towards competitive advantage

Management theory assures that brands can gain competitiveness by implementing differentiation or price strategy (Porter, 1989). Increasingly, real-time engagement and communication contributes distinctive value

to personalised consumer experience and supports dynamic decision-making (Chan & Guillet, 2011; Neuhofer et al., 2012). Brands should enhance their responsiveness by constantly listening, learning and adapting to consumer feedback in real time to optimise outcomes (Sawhney et al., 2005). Brands that are able to develop innovative techniques to manage real-time big data-driven strategies will be able to develop better value for their stakeholders and enhance their competitiveness (Forrester, 2017; Lieb, 2014; Tuškej & Podnar, 2018). Figure 2 demonstrates that time and responsiveness in offering instant gratification is emerging as the third source of competitive advantage.

Classic strategic management assures that competitive advantage can be achieved through price and product differentiation (Porter, 1989). Increasingly "real-time" is instrumental in the experience enhancement process by dynamically understanding the ever-changing consumer needs and drive to exceed expectations through instant gratification (Pretzel Crisps, 2017 and Sullivan, 2011). Brands in the future will be successful only when they have the solution readily available for consumers before they even realise that they have a problem or a need.

Following all contextual information and proactively making alternative arrangements require agility, interoperability with all partners and smart tourism solutions to dynamically co-create itineraries and support customer needs (Buhalis & Amaranggana, 2015; Buhalis & Foerste, 2015; Boes, Buhalis, & Inversini, 2016). Achieving real-time service competitive advantage is based on understanding contextual factors before competitors. These factors include: real-time, co-creation,

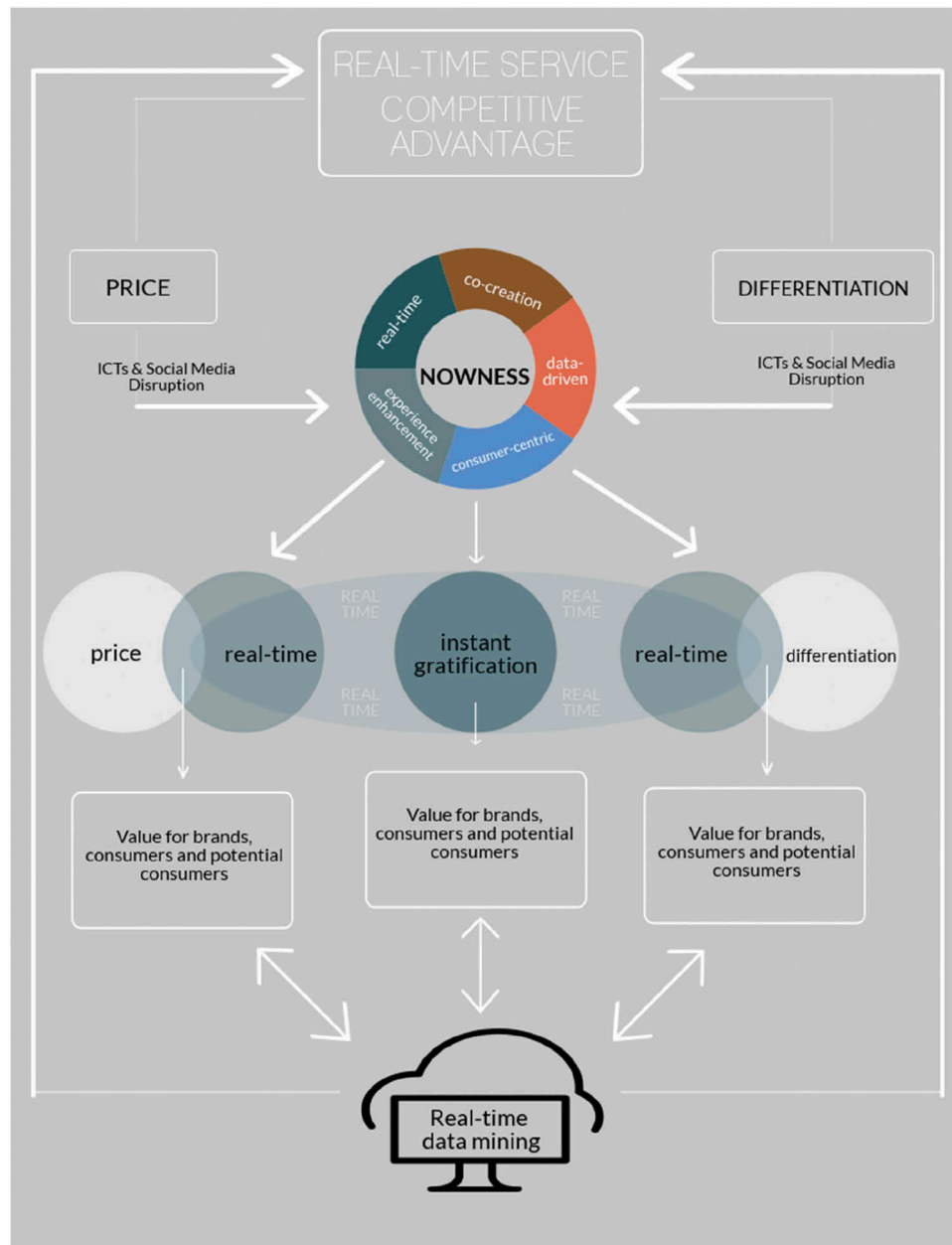


Figure 2. Conceptualisation of real-time service competitive advantage.

Source: Sinarta and Buhalis (2018)

data-driven, consumer-centric, and consumer experience enhancement. These are catalysts for dynamic, flexible, responsive performance for both brands and consumers. ICTs will be utilised within an agile and smart ecosystem, where brands are able to analyse and recognise unique needs of each individual dynamically, and tailor their service through real-time data mining to gain mutually beneficial value instantly. Competitive advantage can be sustained by providing intangible attributes, innovation and high quality of service, by being the fastest service provider (Hunt, 2012). Only those who can provide compelling and

rich experiences to consumers can stay competitive longer term (Boes et al., 2016; Mistilis, Buhalis, & Gretzel, 2014; Pine & Gilmore, 1999).

Nowness and the ability to co-create real-time service emerge as a strong source of competitiveness. Lieb et al. (2013) argue that agile service offerings and time flexibility in service development are critical for brand competitiveness. Real-time data-driven strategies will enable brands to develop competitive advantage and enhance their offering (Forrester, 2017). Consumers are receptive to engagement which is related to their needs at that particular moment. A time-constrained,

context-based strategy is the most pertinent method to tackle constantly-changing consumer demand. Providing service offerings that are highly relevant, based on real-time contexts or situations gives benefits to brands in gaining competitive advantage and amplifying revenue, through dynamic pricing strategies and real-time yield management (Buhalis & Foerste, 2015).

Real-time marketing meets the expectations of consumers for instant, authentic and effortless connection with product and service providers. From the perspective of travel and tourism, consumers utilise social media to gain inspiration, collect and compare information, find desired product or services, share their experience, express feelings, provide input/output on the decision-making process or inspire others (Fotis et al., 2011; Hudson & Thal, 2013; Leung, Law, van Hoof, & Buhalis, 2013; Xiang & Gretzel, 2010). Table 2 illustrates real-time service examples that allow consumers to enhance their engagement at the different stages of their travel experience.

Real-time co-creation and nowness service primarily refer to the during-the-trip stage (Fotis et al., 2011; Neuhofer et al., 2012). Constant connectivity and the ubiquity of digital devices bring interactivity to another level. Consumers require their needs to be acknowledged, recognised and instantly gratified. Using digital services and a wide range of social media they expect brands to satisfy their needs “now” (Patel, 2014). They also require effortless access to real-time information, the development of personalised itineraries and suggestions, as well as alternatives if travel is interrupted (Buhalis & Amaranggana, 2015). This is particularly the case when there are disruptions due to unplanned events, such as emergencies, extreme weather conditions, terrorist attacks or unpredictable travel disruptions. Social media are instrumental for consumers to engage in dynamic conversations with brands. In cases of unexpected events, emergencies and crises customers increasingly require instant service and hold brands accountable for how they handle the situation. For example, if flights are can-

celled they expect convenient rearrangements to minimise disruption to itineraries. A timely response and instant service recovery are expected as well as communication through a range of social media channels (Buhalis & Foerste, 2015; Hudson & Thal, 2013). Interestingly, if one communication channel is overloaded, consumers have no patience to wait and often instantly move to another channel to get an instant response. If call centres are overloaded and there is a long waiting time, they tend to move to Twitter or Facebook or WhatsApp communications in their attempt to get a response to their message. Similarly, they may multiply their message, compliment or complaint across a range of social media to raise their point and trigger a response from brands.

Brands developed the ability to connect with consumers instantly, to personalise products and services and provide real-time engagement across all social media channels around the clock (Lieb, 2014). Online reputation and reviews and the use of reputational management systems are of great importance for increased sales, market competitiveness and therefore overall performance (Viglia, Minazzi, & Buhalis, 2016). This is particularly the case for larger organisations and global brands such as airlines and hotel chains, whilst smaller brands are less pressurised. Consumers expect their preferred brands to care for them, particularly at a time of crisis or distress. Thus, it is paramount for brands to have a good sense of dynamic needs, to show empathy and deliver an adaptive response to consumers' changing demand, in order to be successful. Leadership defines innovation and agility and therefore is critical for the adoption of real-time technologies (Spencer, Buhalis, & Moital, 2012).

Methodology

To conceptually develop the model an exploratory research was conducted to examine how brands develop their technology and social media in order to engage with consumers in real-time and perform

Table 2. Examples of real-time engagement and service.

Social Media Motivations	Example of real-time service (consumer-brand) motivation	Travel Stage
Search for information, look for price & availabilities, product/service features	Consumers would like to reach brands directly through social media channels or digital platforms to get specific information instantly. Brands need to be able to interact, guide and address inquiries online and offline by providing deep links to specific content that is context informed.	Before, during
Self-expression and feedback	Consumers may share their feelings and emotions, both positive and negative, via social media right away, even when the activity is still happening. Brands need to be able to scan the social media sphere, pick up signals and engage in a dialogue, acknowledging compliments and addressing complaints before they escalate. Adding value, addressing service failure and enhancing the experience is of essence in this process.	Before, during, after
Consumption and Post-consumption evaluation	Consumers often share their positive or negative experience as a review. This influences the reputation of the brand dramatically. Consumers may seek acknowledgement, solution or compensation from brands or to notify other consumers (peer-to-peer) in real-time and it is the manner that brands respond that influence their competitiveness.	During, after

Source: Adapted from Fotis et al. (2011)

nowness service. As an exploratory study, qualitative methods were selected (Braun & Clarke, 2013) with an inductive enquiry that started from observation to the development of theory (Farquhar, 2012). A combination of secondary and primary data examined the concept of *real-time service*. The analysis of literature and online information provided the best-practice in real-time marketing. Best-practice digital marketing from industry websites and reports were also consulted due to the novelty of the topic. This offered a good insight into global *real-time service* best practice and helped to formulate a holistic perspective in real-time marketing practice towards generating theory.

The first stage of the primary data collection included a systematic analysis of real-time incidents on social media. This showed a wealth of real-time engagement interactions between brands and consumers. Twitter and Facebook were the two social media channels examined. The filtering of data process took three steps: 1) monitor 20 brand social media channels (e.g. British Airways, KLM, Marriott Hotels, Go Ape, JetBlue, Prêt-a-Manger, Visit Scotland, Hilton Hotels) for a month to capture real-time service in practice. These brands were purposefully selected as best in practice and represented the range of enterprises in tourism and hospitality. 2) employ a hashtag search by brands' name, such as: #Marriott or #BritishAirways, 3) review 115 Twitter and Facebook posts that had a response time of less than 24 h to capture the (near) real-time interaction between consumers and brands.

To analyse the motivation, sentiment and value of real-time interaction captured a codebook was developed. Factors such as the immediacy of the reply, the sentiment of the text or images, motivation of engagement and the value co-created were analysed systematically. Initial codes were added to each data sample to create a pattern which would lead to clearer data categorisation and themes (Saldana, 2016). In the process of reviewing each data set, a total of 31 posts were omitted from data analysis, due to irrelevant context or inappropriate language. Hence, out of 115 posts which were filtered in the initial stages a total of 84 were used for further analysis (72 tweets and 12 Facebook posts) as they met the pre-defined inclusion characteristic. An in-depth analysis of each social media post was conducted. Data saturation was gradually evident after the 48th post, which was shown through recurrence of prominent patterns and motives. The content analysis was conducted using Microsoft Excel, to do a systematic data collection, employ filter options and better categorisation of each theme occurred.

Marriott MLive suite was selected for an in-depth case study using direct observation and a semi-structured interview. This case study enables the researchers to explore, understand and appreciate the real-time global best practice. It was used to generate theory from the pattern of the data (Farquhar, 2012). Purposive sampling was conducted and Marriott Hotels MLive was chosen as a unit of analysis. There were four main considerations to choose one best-practice company which has successfully implemented real-time service to enhance consumer experience: 1) Marriott International Hotels are leading in real-time service strategy, 2) Marriott has a dedicated real-time command centre (MLive) in four countries to engage with its worldwide audience around the clock, 3) Marriott utilises real-time intelligence with a core objective of "Return on Engagement" and co-creation (experience enhancement), 4) Marriott received a gold award for Social Data and a silver award for the Use of Real-time Data categories at the Cannes International Festival of Creativity in 2017 (Marriott, 2016, 2017). Data collection and semi-structured, open-ended interviews were carried out with two senior executives. Direct observation at the MLive was conducted after the interview to have better understanding on how Marriott monitors social media and handles events and posts in real-time. Upon the completion of the interview transcription, an initial analysis of the interview data was carried out to underline emerging themes before conducting an in-depth content analysis (Gribich, 2007). The interview textual data and the analysis of direct observation were filtered and reviewed further to address study objectives. Yet, a single case study carries weaknesses, such as the contribution and credibility of findings (Yin, 2014). Therefore, secondary multiple-case studies were implemented in the third research stage to achieve more credible findings.

Finally, a multiple case study was carried out through online passive observation and document analysis to compare and contrast real-time service cases (Farquhar, 2012). A purposive sampling method was chosen to select appropriate best-practice companies which employ real-time service in their strategy. Four chosen companies, from various industries that allowed variety for the cross-case analysis (Miles, Huberman & Saldana, 2014). The four companies chosen included KLM Dutch Airlines, Visit Philadelphia, Red Roof Inn and Pretzel Crisps. They met the eligibility criteria: 1) Company uses social media and technology as part of online strategy, 2) Company actively engages with consumers using social media or technology in real-time, 3) Company employs immediacy or real-time attributes. Secondary sources were collated from company

websites, annual reports, online publications, public interviews and social media channels. The analysis of data was employed in two stages: an individual analysis of each case study and data integration for the cross-case analysis.

The case studies selection identified companies that have global leadership in different aspects of real-time service. The Dutch Airlines KLM use social media and real-time engagement as a customer service tool and also as a method to manage crises. Visit Philadelphia as a destination management organisation offers real-time surprise and delight experiences with a combination of virtual and physical approaches Visit Philadelphia, 2016. The hotel chain Red Roof Inn is renowned for their highly contextual and personalised advertisements Mobile Marketing Association (MMA) (2014). Finally, Pretzel Crisps aim to engage customers through brand enhancement by real-time listening, surprise and by delighting customers by analysing keywords Pretzel Crisps (2017). The synthesis of all findings, from the 84 real-time incidents on social media of 20 brands, the in-depth Marriott MLive case study, and the multiple real-time service case studies from KLM Dutch Airlines, Visit Philadelphia, Red Roof Inn and Pretzel Crisps provided an evidence-rich basis for developing a conceptual framework to explore real-time co-creation and nowness service and identifying key catalysts.

FINDINGS: real-time co-creation and nowness service as a disruptive force

Findings from each research stage, namely: the social media real-time incidents, the in-depth Marriott MLive case study and the multiple case studies were analysed to demonstrate their contribution to real-time co-creation and nowness service. Then, they are synthesised to support conceptualisation of the real-time co-creation and nowness service ecosystem in Figure 3.

Consumer and brand dynamic interaction and engagement

The findings from the systematic analysis of real-time incidents on social media reveal that consumers use social media to express their feelings, find information about products and services, as well as to co-create value and experience with brands. Social media is often used for expression of feelings, in anticipation for a reaction in real-time. This is a major motivation for engagement. Various sentiments are evident, comprised of positive, negative and neutral interactions. When consumers feel satisfied, they would like to express their appreciation and praise brands on social

media in real-time. This generated organic User Generated Content (UGC) and positive online Word-of-Mouth (e-WOM) for brands, encouraging potential new customers. Frequent positive engagement turns customers into advocates and brand ambassadors.

Real-time incidents on social media illustrate the power of engagement

Consumers express their feelings by sharing both positive and negative feelings. *They express their emotions, appreciation and possibly dissatisfaction. For example, a guest at the AC Hotel by Marriott Irvine expressed the sentiment "Loved the European style breakfast at @achotelirvine, and will come back because all the staff is fabulous!". Often consumers approach brands to seek particular information about brand's service or to enhance their experience. A passenger was asking @British_Airways "how many hours before my flight can I use the Concorde Room in Singapore?" Real-time engagement in a catalyst in co-creation of value and managing experiences. For example, a guest twitted the Yeouido Park Centre, Seoul – Marriott Executive Apartments "@MarriottintlYeouido can you get me tickets for the Nanta show tonight and what is a nice place for Korean food close to the theatre? #Seoul."* Dynamic interaction enables brands to offer individual and personalised service based on contextual information. Consumers interact digitally to create and enhance their own experience in real time.

A high volume of information is therefore received by consumers on a daily basis. Real-time communications and feedback from consumers are effectively touchstones for brand product and service improvement. Consumers hashtag brand names or tag brands to engage brands through social media and expect instant customer service and feedback. The range of information sought by consumers usually refers to a brand's specific product or service information. They may reconfirm uncertain information, seek explanation about real-time cases or request for particular facilitation. Evidence reveals that consumers also use social media to complain and notify brands about unsatisfactory facilities "@JetBlue my tv is always broken on your flights." If unsatisfactory service is received on one communication channel, such as call centres, email or physical encounters consumers often escalate the issue across multiple channels. For example, one consumer repeated a question to a brand via Twitter which he had previously asked via call centre but did not get a sufficient answer. He then expressed dissatisfaction on service previously received: *"Ok thanks can I recommend you retrain your call centre staff as they appeared confused on the matter and said the rules changed in June."* Evidently consumers also utilise social

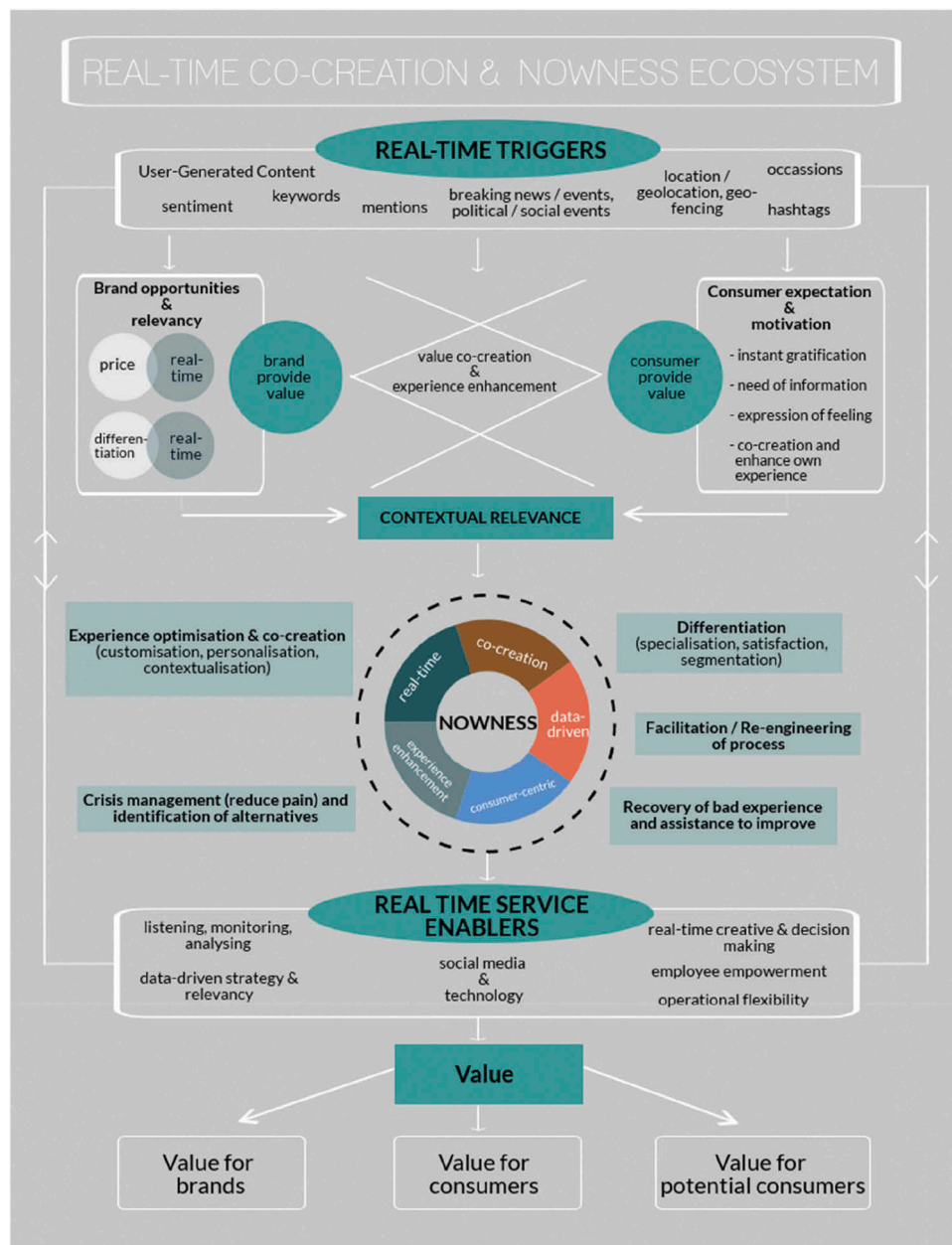


Figure 3. Conceptualisation of real-time co-creation and nowness service ecosystem.

media to provide suggestions or notify brands when a particular feature of service did not meet their expectation. “@MarriottIntl why can’t I download a receipt on the website?!? You are NOT business friendly. Going to @HiltonHotels.” His unsatisfactory feeling about lack of expected service from brand leads him to lose his sense of loyalty. Consumers expecting brands to solve their issues in real-time and expose them in an attempt to force them to take action. The real-time data shared by consumers about their experience or feeling, whether it is positive or negative, are valuable sources to rectify overlooked problems and develop even better service offerings for consumers in the future. More importantly,

consumers expect that organisations can take immediate action and effectively create a public dialogue with the brand, which is often watched by many others, affecting their reputation.

The content analysis illustrates that real-time interactive interactions empower consumers to contribute and receive feedback, as well as to share their views and preferences. As consumers are always connected, they require fast solutions and they do that on the public domain. Often organisations are proactive at fetching information from the external context and communicate it to users at the right time. Examples include, time and location dependent offers from

restaurants or alerts from transportation companies when disruptions or delays happen. Both brands and consumers gain mutually beneficial value, such as positive advocacy, real-time service improvement, loyalty, sales, instant gratification and experience enhancement. There are plenty of examples of interactive engagement between brands and consumers through social media. Dynamic interactions provide extra value to consumers that often offer positive advocacy back *"Thank you @Marriott and @RenHotels not what we did to deserve this special treat from @CherylsDesserts #surprise #fun #treat"* or express their brand loyalty publicly *"Thank you for the reply – I did enjoy a lovely glass of wine on my flight to Helsinki. Will be flying only KLM from now on Xx."*

Instant gratification can lead to sales facilitation and conversion. For example, a consumer is asking *"@KLM trying to book a round-trip. My daughter is less than 2 in outbound and more than 2 in inbound. We can't process booking online. Help?"* The airline replied in 25 minutes: *"Hello, allow us to help create that booking for you! In order to do so please send a Direct Message so we can collect the needed details."* It can also facilitate the customer experience enhancement *"Great experience with @KLM flight 0862 Complimentary upgrade, champagne and Dutch houses to celebrate our engagement!!"* Product and service improvement may emerge from real-time feedback *"Ok thanks can I recommend you retrain your call centre staff as they appeared confused on the matter and said the rules changed in June."* Inteaction can lead to conversion to sales, co-creation and service enhancement *"@1Bomboloni can I book a table for 2 tomorrow at 19.30? It's their birthday so can I sneak in a cake too? Tried to call but no answer ?"* with Brand replying within 2 h: *"done, see you then."*

Marriott MLive: data-powered real-time co-creation and nowness service

The research illustrates that the best innovator in real-time service is Marriott Hotels and their Marriott MLive suite. Through four control centres around the world, they engage with consumers in real time to enhance their experience (Table 3). They then personalise and contextualise experiences through dynamic co-creation with stakeholders. Nowness service suggests that brands strive to ensure that the personalisation of service is delivered based on specific needs and preferences at that particular moment. The case study illustrates that flexibility is required for the ever-changing consumer demand. An adaptive approach is critical to create authentic engagement.

Table 3. Examples of Marriott MLive real-time service scenarios.

Scenarios	Strategy action	Key Performance Indicators
MLive finds a story about a backpacker from New Zealand traveling the world with his Lego figurines and posting their experience on social media. Marriott contacted the Lego backpacker and offers to host him at the AC Hotel Ciudad de Sevilla.	Enabling advocacy and word-of-mouth	This action has resulted in a fresh content for the AC hotel and the additional 370,000 social followers joining the conversation.
A traveller asked for a hotel recommendation on Twitter for Nashville Girls' weekend. MLive noticed (despite not mentioned or tagged) and promptly gave relevant hotel advice. In addition, it shared content regarding local nightlife add value and enhance the experience.	Promotion, problem solving, distribution	Within 13 minutes of the conversation, the traveller posted on Twitter sharing that she had booked a room with Marriott.
MLive identifies a tweet from a guest at the Domes of Elounda Autograph Collection Hotel in Greece. The team exceeded the guest's expectation by creating on-property breakfast moment and granting Marriott Rewards points as a surprise.	Augmenting the experience towards word-of-mouth	The guest shares his satisfaction on Twitter and produce positive user generated content for the Autograph Collection Brand and hotel.
MLive jumped into perfect opportunity to utilise the Pokémon Go craze around the world by engaging Nick Johnson, the first gamer to catch all 142 Pokémon within two weeks in the United States, right before he becomes the worldwide trending topic as a result of his achievement. Marriott offered him to travel around the world to catch the remaining rare Pokémon in Asia, Australia and Europe.	Taking advantage of contemporary trend to gain engagement and advocacy	- Earned organic PR coverage worldwide, including news coverage by USA Today, CNN, Buzzfeed, Business Insider and Adweek. - More than 800 global news outlets covering the story, 281 million of total PR impressions. - Driving brand buzz across social media channels (increased by 40%) with around 9 million global Twitter impressions. - Earned Gold Distinction in real time response from the 9th Annual Shorty Awards 2016

The findings indicate that social media, digital channels and mobile applications, can provide consumers with immediate response and instant gratification from brands which underpin the co-creation process. One of the interviewees emphasised the importance of co-creating value in a manner that consumers can feel the authenticity of the engagement:

Social is not just about 1 to 1 interaction, it's also about that kind of content strategy that you see the opportunities as they emerge, the trends as they evolve and start to say what can we do, how can we be involved in that as a brand, without really being seen as paid for.

The Marriott case study revealed that social media engagement is not just a way to create a dialogue with consumers to develop brand awareness. Marriott puts consumers at the centre of their stories, as a facilitator of the enhanced moment creation. Consumers become prosumers and advocates of brands they believe in. Brands can gain competitiveness and engage with the community to achieve a twofold loyalty and a long-lasting relationship.

Best practice in real-time co-creation and nowness service

Findings from multiple case studies reveal that the four companies, i.e. KLM, Visit Philly, Pretzel Crisps and Red Roof Inn have the ability to recognise consumer needs proactively and co-create value with them in real-time. These brands target influential consumers to enhance their experience and add value in near real-time. This benefits brands with authentic interactions, organic PR coverage, service recovery, conversion to sales, and enhance their competitive advantage. The main similarities among these four brands are their dynamic, engaging interaction with consumers and how they personalise and contextualise their service offerings.

KLM is committed to respond to all online enquiries via social media, within 1 h, 24 h a day. Visit Philadelphia and Pretzel Crisps use *surprise and delight* to enhance consumer's experience, by integrating physical and virtual interactions. Red Roof Inns are able to convert crisis situations into a profitable business by identifying their consumers' critical needs before other competitors do. They provide a real-time solution to consumers' demand at the exact time they need it, using the channels that consumers actively engage with.

Scanning all media in real-time enables brands to improve their agility and relevance as well as managing customer experience and value in real-time. By listening to consumers and stakeholders in real time and also by scanning the context in geo-fenced areas brands can

take instant action to proactively ameliorate nowness service and co-create valuable experiences in real-time. Consumers get an answer or a positive surprise "now," co-creating their experience in real time. This practice not only supports profitability but also supports competitiveness through nowness service.

Synthesis: real-time co-creation and nowness service ecosystem

The conceptual framework of *real-time* service ecosystem illustrated in [Figure 3](#) amalgamates different elements elaborated in the findings section (consumer and brand dynamic interactions, real-time triggers, advantages of real-time service, real-time service enablers and characteristics). The conceptualisation of real-time co-creation and nowness service ecosystem illustrates how experiences can be further enhanced, generate added value leading to contextual relevance. Activating nowness service can generate advantages of real-time co-creation. To ensure that nowness service is achieved several real-time service enablers are identified leading to value for brands, consumers and potential customers.

Real-time experience triggers towards contextual relevance

The analysis of the documents, interviews and direct observation as well as the Marriott case study have revealed eight real-time experience triggers which are presented in [Table 4](#). These triggers effectively initiate the contextualisation process and prompt brands to engage with consumers' requests, compliments, complaints and important or viral news. These triggers vary from UGC by influencers, sentiments, specific keywords or hashtags related to brands or events, brand mentions, to breaking news or real-time events which are relevant to brands and their consumers. Although the triggers in the case of Marriott MLive are mainly capitalising real-time data, based on geolocation and geo-fencing, Marriott Hotels and other brands still utilise additional real-time triggers, that emerge from the ecosystem through any signal. For example, employee observations, face to face and telephone interactions with consumers, engagement with social media and review sites may spark an incident and trigger a response to ensure comprehensive coverage and relevance. The triggers generate nowness service and influencer collaboration opportunities, which help brands to co-create experiences online and strengthen their brand competitiveness.

Real-time co-creation and nowness service are expected to be relevant for both parties and support value and experience enhancement. Brands can

Table 4. Real-time co-creation triggers for nowness service.

Real-time triggers	Implications
User-Generated Content	Consumers regularly share user generated content about brands, often as things happen. Conversation on line with peers then often emerge about particular experiences with brands. Marriott Hotels implemented alerts to identify influencers in social media and explore the conversation about their brand in real time. Depending on the conversation they often engage proactively with consumers, especially with ones that have many followers, have engaging and favourable content.
Location/geolocation & geo-fencing	Smart devices allow brands to provide geographically relevant content, posted at specific locations at the time that customers visit. Geo-fencing is critical to understand contextual factors in real-time and trigger real-time response. It can also be used for push notifications of special geo and time-based offers and promotions.
Sentiment	Consumers share positive or negative opinions, emotions and feelings in social media. Analysing message sentiment based on the tone or particular words used can help brands develop their engagement. Praise is often acknowledged and encouraged towards creating advocates. Criticism and complaints are addressed instantly, often off line or in private conversations, to improve experience and regain confidence in real-time. This can prevent the escalation of complains and service recovery. Protecting the brand integrity and reputation online is of paramount importance as this is a major asset for tourism and hospitality organisations.
Keywords	Leveraging keywords by tracking brand names, locations or key attributes can also assist brands to engage with consumers and to capture opportunities to cocreate value. Events or related news are good engagement opportunities when people discuss a theme, location, activity or a topic.
Mentions Tags	When consumers mention or tag brands, they want to engage with brands intentionally. This is to share experiences or feelings, ask assistance or information, complain, or simply engage the brands. Acknowledging, responding and supporting are critical to support customers, maintain engagement and improve experience.
Hashtags	Hashtags track particular topics for specific conversations. Hashtag tracking is used to track particular event, topic or campaign. Examples include #Marriott #Bournemouth #thailand #royalwedding #pokemon #tennis #brexit #christmas #olympics. It allows tourism companies to theme activities and target special interest tourism or relate to an event.
Breaking news or events, Political or regular social events	Break news Incidents may change the context dramatically and therefore they must be served through real-time interaction. Real-time monitoring of news assist brands to update stakeholders by providing reliable information. This may include severe weather, natural disasters, earthquakes, storms, strikes, political unrests, violence outbreaks, accidents, terrorism attacks and other crises and unpredicted eventualities. Real-time knowledge about a particular crisis or negative situation may be critical for customer safety and also for the decision whether to travel to a destination or not. Brands have a duty of care to their customers and also need to protect employees and assets. Engaging with real time, accurate and reliable information is critical in a crisis situation. It can also avoid embarrassment when brands have pre-programmed content that may be inappropriate given the current situation. For example promoting holidays at a destination when there is a natural disaster and there is a need for humanitarian aid is inappropriate and embarrassing. Viral stories can also be effective to support brands to achieve authentic engagement and advocacy. For example, utilising positive events, such as the royal wedding for promotion and themed campaigns can gain exposure and bookings for the company. Real-time decision-making is required in seizing the opportunity to do so.
Occasions	Consumer occasions include special memorable moments that consumers share in the public domain that are related to a tourism organisation. When brands proactively acknowledge consumers' important occasions (e.g. birthday or special celebrations), they can offer a surprise and delight real-time experience that leads to positive advocacies and conversations.

Source: Adapted from Sinarta and Buhalis (2018)

optimise the service by providing timely actions, complementing the price and differentiation strategy, and exchange value with consumers. Brands proactively identify information and utilise real-time triggers to co-create value. They enhance consumers experience in real-time, by exchanging mutually beneficial values based on real-time contextual and situational relevance.

Brands need to dynamically explore the situational and contextual factors (external context) in real-time for both brands and consumers. Blending this with consumers' individual needs (internal context) can add value. For example, a culinary brand may identify a negative sentiment from a person who shares his frustration about finding food in the middle of the night on social media (UGC). The brand may decide to take action and provide consumer value, knowing that this person stays in close proximity (geolocation), by identifying this person's possible preference from his public social media profile (internal context) and choosing to *surprise & delight* him with proposing warm food suitable for

rainy and cold nights (external context). The interactions between consumers and brands reflect the dynamic relationship, where both parties continually and reciprocally exchange value that is contextually relevant. The relationship value is mutually advantageous in every interaction. Co-creation of value happens by exchanging resources and information in real team leading to contextually relevant nowness service.

Nowness service and advantages of real-time co-creation

Nowness service emerges through the ability of engaging with consumers dynamically at the present time both physically and digitally. "The service of now" or "nowness" reflects the immediacy nature and the agility of brand's performance towards leveraging presence. Nowness leads a paradigm shift to a new era where new agile practices are performed in real-time to maximise the value co-created for all stakeholders within a smart and

agile ecosystem. These practices are consumer-centric, personalised and contextualised and proactive and reactive in nature. Nowness disrupts the industry by facilitating the agility of brands to co-create experiences dynamically. Those organisations that achieve nowness are vibrant value co-creators, whilst the rest of the market is seen as irrelevant and obsolete. Nowness service is based on five interconnected characteristics, namely: real-time, co-creation, data-driven, consumer-centric and experience enhancement as illustrated in Table 5.

The core values of nowness: *real-time, co-creation, data-driven, consumer-centric and experience enhancement* aim to elevate consumer's experience and deliver hedonic gratification, exceed expectations and create memorable experiences. The environment of nowness is very dynamic and agile, incorporating various scenarios from highly positive (experience optimisation) to very negative (crisis management). It can be performed in virtual or physical environments, public or private, online or offline. Brands can empower consumers with resources to co-create their own experience through collaboration and active engagement with relevant audiences. Brands need to be able to identify different consumer personas based on particular contexts and customer Intelligence. For example, using a combination of *historical data* (such as purchase history or previous relevant interactions), *current data* (such as consumer's social profile or real-time updates), and *predictive data* (analysing historical behaviour and current context) to provide personalised service offerings. This process drives better segmentation and identification of potential needs and requirements. Specific services can be co-created for

football fans, divers or tennis players. Ultimately, this process optimises the value co-created for each stakeholder.

The case studies illustrate that the ability to integrate real-time knowledge of consumer demand, geo-data, contextual elements, online search behaviour and other relevant factors add higher value to consumer-brand engagement. Table 6 reveals the five advantages of real-time service and the value created in each category emerging from the multiple case studies. Experience optimisation and co-creation through perhaps personalised surprise and delight can make consumers feel special. Differentiation through providing specially personalised recommendations within the customer location and using contextual information in real-time can revolutionise experience. Enabling and facilitating service, by providing high quality 24/7 service, using social channels, can improve conversion and customer spent. In cases of service failure or unexpected events, such as adverse weather recovery or flight delays), brands can minimise the impacts and reduce bad experience by demonstrating that they care.

Perhaps most importantly at a time of emergencies and crisis management, real-time communication services are essential, as tourists are by definition visiting hostile and unfamiliar environments. Often these environments are natural, unprotected and prone to risk. These may include natural disasters (such as earthquakes, storms, tornados, cyclones, hurricanes, wildfires, avalanches, tsunamis, volcanoes, landslides, flooding) or social disasters (where community behaviour breaks down, for example, in order of increasing magnitude such as demonstrations; stampedes;

Table 5. Interconnected characteristics of real-time co-creation and nowness service.

Real-time	- Immediate and relevant - Instant engagement - Dynamic environments and delivery types - Virtual or physical environments - Public or private engagement - Online or offline interactions
Co-creation	- Value co-creation, active engagement with consumers - Brands empower consumers to engage dynamically - Resources to co-create own experience through collaboration - Brands as a facilitator of service re-engineering - Instant gratification
Data-driven	- Tailored products and services - Uniquely based on consumer's needs - External context aware - Personalisation and contextualisation - Differentiation of service offerings in real-time
Consumer-centric	- Dynamic pricing and revenue management in real-time - Identification of different consumer's persona based on particular context - Customer Intelligence based on - historical data (- purchase history or previous relevant interactions) - current data - consumer's social profile or real-time updates - predictive data - pattern recognition and forecasting of choice
Experience enhancement	- Focus on real-time experience enhancement - Creating utilitarian or functional gratification for consumers - Deliver hedonic gratification - Exceed expectation and co-create memorable experience

Table 6. Advantages of real-time co-creation.

Advantages of real-time	Characteristics and Implementation	Value for stakeholders
Experience optimisation & Co-Creation (customisation, personalisation, contextualisation)	Brands implement active listening Dynamic interaction with consumers Co-create unique personalisable and memorable experiences Manage sentiment dynamical Engage with consumers proactively Provide contextually relevant value Empathy and authentic engagement drives increasing value Support consumers through active engagement Combine interaction at all touch points-both digital and physical Personalise experience Addressing individual needs, abilities and preferences	Consumer Satisfaction Enhanced experience Create advocates Brand loyalty Positive e-WOM Brand awareness Increase sales Differentiation of service Enhanced experience Consumer satisfaction Positive e-WOM Organic Engagement Authentic User Generated Content Convenience for consumer Enhanced experience Expectation management Complaint management Instant gratification Proximity/location-based offerings Consumer engagement
Differentiation (Specialisation, Satisfaction, segmentation)	Expectation Management Active/responsive interaction with consumers 24/ Help consumers out of normal hours Context drives service delivery Use the communication channels that consumers prefer Real-time data mining for product & service enhancement Increase predictability from frequent occurring instances Re-engineer processes & service encounters. Real-time data analysis of emerging issues Capture real-time incidents and response Manage consumer's itinerary & support online behaviour Geo-tagging and location-based recovery mechanism Targeted queries provide highly relevant & accurate solutions -Alert/investigations/improvement/change in real-time	Proactive service provision Problem-solving Complaint handling Service recovery Consumer satisfaction Creating advocates Demonstrate that you care Manage crisis and emergency Reduce suffering Increase trust Support perceived value Improved experience Instant response Reduce potential casualties Reduce extra spent
Facilitation/Re-engineering of process	Use of social media to provide critical information in real-time Manage emergencies and provide response in real-time Utilisation of keywords and sentiment analysis Identify alternative plans in real-time Manage crisis situations and mitigate experience Reduce impact and manage effect for brands	
Recovery of bad experience and assistance to improve experience		
Crisis management (reduce pain) and identification of alternatives		

riots; terrorism; conflicts; or even wars) or technological disasters (e.g. industrial and technological incidents involving explosions or chemical and radiation emissions; accidents during the transportation of humans or hazardous materials, structural failures of bridges, buildings, power lines, dams or mines, or vehicle and train accidents). Often one disaster triggers another one and a complex set of failures occurs, with economic, social, physical and environmental consequences, all within a state of insecurity (Reed, 2011). These events may have devastating effects on tourists that are caught out in these occurrences, away from home and familiar support mechanisms. Real-time communication during crisis management is critical for minimising the impacts and consequences of disasters, lead people to safety and recover the experience within a responsibility and duty of care framework.

The analysis of engagement from the case studies shows that ICTs can provide a range of real-time advantages for brands. Responding to negative enquiries or complaints timely, by providing explanations, can prevent escalation of issues. The consequences of a service breakdown can be minimised as issues can be identified and addressed instantly. Agile organisations intervene and act in real-time to ensure the best possible outcome for all stakeholders. This helps to quickly understand contextual situations, pull different resources, mobilise stakeholders to address consumers' needs and to effortlessly achieving those needs. Using Machine Learning (ML) and Artificial Intelligence (AI) through big data mining can instantly develop a warning system and support decision-making.

Real-time service enablers

To facilitate nowness service and real-time agility requires brands to utilise several essential external and

internal enablers, as illustrated in Table 7. External enablers include brand's Social media, technology and online presence, the utilisation of analytic, listening and monitoring tools as well as the implementation of data-driven strategy. Organisations need to effectively develop radars to constantly monitor all aspects of conversations, both on social media, and with face to face interactions through an agile system. All positive and negative feedback, as well as critical information affecting performance, need to be recorded and actioned in real-time. Instant analysis and decision-making support problem solving, differentiation, contextualisation and co-creation of value towards instant gratification.

External enablers consist of content curation and decision-making in real time as well as the empowerment of employees through a flexible operational structure. Once the agile system identifies real-time issues to be addressed, internal enablers can support prompt decision ameliorating situations. For example, airline systems may allocate travellers to the next available flights when a flight is missed or cancelled due to adverse weather conditions. Similarly, when the agile system identifies a signal that can support value enhancement, analysis should lead to nowness service that enhances the co-created value for all stakeholders. The identification of the birthday celebration of an in-house hotel guest, for example, should trigger a surprise, with cake or a special treat. The expression of a complaint about cleanliness should offer an apology followed by the dispatch of the housekeeping team. The expression of a compliment should trigger an acknowledgement and perhaps a treat to the satisfied consumer and the praised member of staff. Positive engagement can support instant gratification, enhanced value co-creation, loyalty and advocacy.

Table 7. Real-time service enablers.

Essential	
External enablers	Social media, technology and online presence: Interconnected social channels (e.g. Facebook, Twitter, Instagram, Google+, Snapchat, etc.), interactive and responsive websites, and/or brand's mobile apps. Listening, monitoring and analysing tools: Developing monitoring and search tools in social media through Twitter's Advanced Search, Facebook for Developers, and other third-party tools such as, Hootsuite, Google Alerts, Sprout Social, Mention, Keyhole, Netbase, etc. Data-driven strategy and data relevancy: A strategy for data mining in Real-time where all relevant data for the context of brands and consumers are examined dynamically. A combination of internal context including consumer's individual preferences, abilities, likes and external contextual context such as location or weather can be used to optimise suggestions.
Brand's internal enablers	Real-time creative content curation and decision-making: Real-time content curation can include an Intuitive, quick capture of trends Support both virtual/digital and physical/natural interactions These interactions can be either private or public according to the value co-creation Employees' empowerment: Well-trained, predictive-scenario management (Ability to solve problems instantly and empowering decision-making for individuals Detailed product and service knowledge Human empathy and personable and approachable manner Operational flexibility Highly adaptive to emerging trends (e.g. flexible tactics based on real-time data) Highly responsive to constant changes in consumer behaviour

Systems and staff should be empowered to act instantly and to maximise the value for all stakeholders involved in the ecosystem. This may be done through employee empowerment and operational flexibility. Allocating resources as well as trusting and empowering front-line staff can co-create real-time value for the benefit of all stakeholders.

Achieving nowness service requires a layer of infrastructure using smart technologies that integrate the real-time service ecosystems. These include sensors and beacons, big data management, cloud computing, Decision Support Systems, Artificial Intelligence, and Machine Learning. These hardware and software technologies collect and analyse information dynamically as well as trigger several processes in real time. ICTs act as the catalysts in an agile ecosystem to enhance real-time service performance based on the brand's specific objectives.

Nowness service can be achieved by identifying the enablers to perform real-time co-creation: use social media and the organisational online presence to engage in conversation with various stakeholders; *listening and monitoring* to identify, analyse and filter real-time data that can influence the real-time value co-creation; and *deliver and engage* with real-time content curation through creative and innovative content. Decision-making in real-time through employee empowerment can facilitate seamless and dynamic interaction with all stakeholders on the ecosystem using social media and technology. The feedback loop shows the interrelations of these enablers with real-time triggers. These triggers can perform interchangeably and dynamically, particularly in the listening and monitoring phase, and also evaluate the impact of the real-time co-creation and the value produced, so further actions can be facilitated.

Conclusions

This exploratory study reveals that real-time co-creation and nowness service provide an innovative approach to co-creating value with consumers, across multiple platforms. Interactions occur in real-time, at the exact moment consumers are willing to engage and co-create value with brands. The research shows that real-time service adds value to brands' competitiveness by using real-time big data from consumers, supplemented with contextual information relevant to both parties. Consumers' unprecedented access to information and the constantly changing demand towards a highly personalised service have effectively disrupted and re-engineered the service ecosystem. Nowness service takes advantage of the consumer interactive online engagement with the entire service ecosystem across multiple platforms and media. Brands are able to increase

their performance effectiveness and efficiency by dynamically engaging and performing timely actions based on real-time consumer intelligence and contextual data.

This paper also makes a best practice contribution, as provides immediate support to the wider tourism and hospitality industries and destinations to adopt dynamic and agile methodologies for co-creating real-time value. Other service domains can also benefit from the real-time service concepts and mechanisms to establish dynamic engagement with consumers and offer agile and context-based services. Real-time services can be offered by both large and small businesses. Large companies, such as Marriott Hotels with its dedicated real-time command centre or airlines such as KLM with its 1-h response promise, have already developed strategies to implement that. However, Small, Medium Enterprises (SMEs) can use freely available tools and engage in dialogue with consumers in real-time. Brands can also expand their virtual size by engaging with other service providers and members of the ecosystem to expand their service scope and offer holistic experiences. Future research can benchmark best practice in real-time service and conceptualise how this will revolutionise business and network competitiveness.

Disclosure statement

No potential conflict of interest was reported by the authors.

References

- Baer, J. (2013). *42 Percent of Consumers Complaining in Social Media Expect 60 Minute Response Time* [online]. New York, NY: Convince & Convert. Retrieved from <http://www.convinceandconvert.com/social-media-research/42-percent-of-consumers-complaining-in-social-media-expect-60-minute-response-time/>
- Baghirov, F., Zhang, Y., & Hashim, N. (2019). Facebook fan page management for global airlines. *Tourism Review*. doi:10.1108/TR-03-2018-0045
- Boes, K., Buhalis, D., & Inversini, A. (2016). Smart tourism destinations: Ecosystems for tourism destination competitiveness. *International Journal of Tourism Cities*, 2(2), 108–124.
- Braun, V., & Clarke, V. (2013). *Successful Qualitative Research: A Practical Guide for Beginners*. London: Sage.
- Brogan, C., & Smith, J. (2009). *Trust Agents*. Hoboken, NJ: John Wiley & Sons.
- Buhalis, D., & Amaranggana, A. (2015). *Smart Tourism Destinations Enhancing Tourism Experience through Personalisation of Services* [online]. In I. Tussyadiah & A. Inversini Eds., *Information and Communication Technologies in Tourism 2015*, Lugano, 3–6 February, 2015 (pp. 507–521). Switzerland: Springer.
- Buhalis, D., & Foerste, M. (2015). SoCoMo marketing for travel and tourism: Empowering co-creation of value. *Journal of Destination Marketing & Management*, 4(3), 151–161.

- Buhalis, D., & Law, R. (2008). Progress in Information Technology and Tourism Management: 20 Years on and 10 Years after the Internet – The State of eTourism research. *Tourism Management*, 29, 609–623.
- Chan, N., & Guillet, D. (2011). Investigation of Social Media Marketing: How Does the Hotel Industry in Hong Kong Perform in Marketing on Social Media Websites? *Journal of Travel & Tourism Marketing*, 28(4), 345–368.
- Chandler, J., Salvador, R., & Kim, Y. (2018). Language, brand and speech acts on Twitter. *Journal of Product & Brand Management*, 27(4), 375–384.
- Chandler, J., & Vargo, S. (2011). Contextualization and Value in Context: How Context Frames Exchange. *Marketing Theory*, 11(1), 35–49.
- Christopher, M., Payne, A., & Ballantyne, D. (2002). *Relationship Marketing – Creating Stakeholder Value*. London: Elsevier.
- CMO Council. (2017). *Gaps and Fragmentation Plague Marketers Looking to Deliver Seamless Engagement to Connected Customers*. San Jose: Author. Retrieved from <https://www.cmocouncil.org/media-center/press-releases/5339>
- Costa, C., & Coelho Do Vale, R. (2018). To tell or not to tell? The impact of communicating consumer participation in new product development. *Journal of Product & Brand Management*, 27(2), 158–171.
- Court, D., Elzinga, D., Mulder, S., & Vetvik, O. (2009). *The Consumer Decision Journey*. Dallas: McKinsey & Company. Retrieved from <http://www.mckinsey.com/business-functions/marketing-and-sales/our-insights/the-consumer-decision-journey>
- Darmer, P., & Sundbo, J. (2008). Introduction to Experience Creation. In J. Sundbo & P. Darmer (Eds.), *Creating Experiences in the Experience Economy* (pp. 1–12). Cheltenham: Edward Elgar.
- Deutsch, J. (2014). Guide to Real-time Marketing. *Digital Marketing Magazine*. Retrieved from <http://digitalmarketingmagazine.co.uk/articles/minttwist-s-guide-to-real-time-marketing/1148>
- Dorcic, J., Komsic, J., & Markovic, S. (2019). Mobile technologies and applications towards smart tourism – State of the art. *Tourism Review*, 74, 82–103.
- Econsultancy. (2015). *Q&A: Jeff Rajeck on Real-time Marketing and Personalised Customer Experience*. London: Author. Retrieved from <https://econsultancy.com/blog/66593-q-a-jeff-rajeck-on-real-time-marketing-and-personalised-customer-experiences/>
- Evergage. (2014). *Real-time for the Rest of Us*. Somerville: Author.
- Farquhar, J. (2012). *Case Study Research for Business*. London: Sage.
- Firebelly. (2014). *Real-time Marketing Vs Real-time Response: Why It's Important You Know The Difference*. Indianapolis: Author. Retrieved from <https://www.firebellymarketing.com/2014/11/real-time-marketing-vs-real-time-response.html>
- Forrester. (2017). *The Evolution to Real-time Customer Experience*. Cambridge: Author. Retrieved from http://www.datastax.com/wp-content/uploads/resources/whitepaper/Forrester-CX-TLP_DataStax.pdf
- Fotis, J., Buhalis, D., & Rossides, N. (2012). Social Media Use and Impact during the Holiday Travel Planning Process. In M. Fuch, F. Ricci, & L. Cantoni (Eds.), pp. 13–24. *Information and Communication Technologies in Tourism*, Sweden. Austria: Springer-Verlag Wien.
- Fotis, J., Buhalis, D., & Rossides, N. (2011). Social Media Impact on Holiday Travel Planning: The Case of the Russian and the FSU Markets. *International Journal of Online Marketing*, 1(4), 39–50.
- Garner, R. (2013). *Search and Social: The Definitive Guide to Real-time Content Marketing*. Indianapolis: John Wiley & Sons.
- Girish, D. (2016). *Contextual Marketing: A Contextual Triggers Digital Marketers Must Use Right Away*. Beaconstac. Retrieved from <https://blog.beaconstac.com/2016/05/contextual-marketing-3-contextual-triggers-digital-marketers-must-use-right-away/>
- Gretzel, U., & Fesenmaier, D. (2009). Information Technology: Shaping the Past, Present and Future of Tourism. In T. Jamal & M. Robinson (Eds.), *Handbook of tourism studies* (pp. 558–580). Thousand Oaks, CA: Sage.
- Gribich, C. (2007). *Qualitative Data Analysis: An Introduction*. London: Sage.
- Harwood, T., & Garry, T. (2017). Internet of Things: Understanding trust in techno-service systems. *Journal of Service Management*, 28(3), 442–475.
- Hays, S., Page, S. J., & Buhalis, D. (2013). Social media as a destination marketing tool: its use by national tourism organisations. *Current Issues in Tourism*, 16(3), 211–239. doi:10.1080/13683500.2012.662215
- Heine, C. (2013). Starbucks Pushes 'Snow Day' on Facebook, Twitter: Coffee Giant Uses Clever Real-Time Ad Buy. *Adweek*. Retrieved from <http://www.adweek.com/digital/starbucks-pushes-snow-day-facebook-twitter-147187/>
- Howell, K. (2014). *In the Social Moment: Real-time Social Engagement – Benchmarking Brands in Social Media*. Surrey: Immediate Future (IF).
- Hudson, S., & Thal, K. (2013). The Impact of Social Media on the Consumer Decision Process: Implications for Tourism Marketing. *Journal of Travel & Tourism Marketing*, 30, 156–160.
- Hunt, S. (2012). Trust, Personal Moral Codes, and the Resource-Advantage Theory of Competition: Explaining Productivity, Economic Growth, and Wealth Creation. *Contemporary Economics*, 6(2), 1–90.
- Law, R., Buhalis, D., & Cobanoglu, C. (2014). Progress on Information and Communication Technologies in Hospitality and Tourism. *International Journal of Contemporary Hospitality Management*, 26(5), 727–750.
- Leung, D., Law, R., Van Hoof, H., & Buhalis, D. (2013). Social media in tourism and hospitality: a literature review. *Journal Of Travel & Tourism Marketing*, 30(1–2), 3–22. doi:10.1080/10548408.2013.750919
- Lieb, R. (2014). *Real-time Marketing: The Ability to Leverage Now - Webinar*. Retrieved from <https://www.slideshare.net/lieblink/real-time-marketing-the-ability-to-leverage-now-webinar>
- Lieb, R., Groopman, J., & Li, C. (2013). *Real-Time Marketing: The Agility to Leverage 'Now'*. Altimeter, Boston. <https://www.slideshare.net/Altimeter/report-realtime-marketing-the-agility-to-leverage-now-by-rebecca-lieb-jessica-groopman>
- Marriott. (2016). *Marriott International Launches M Live Europe, the Company's Fourth Global Marketing Real-Time Command Centre*. London: Author. Retrieved from <http://news.marriott.com/2016/07/marriott-international-launches-m-live-europe-companys-fourth-global-marketing-real-time-command-centre/>
- Marriott. (2017). *Marriott International Wins Two Creative Data Cannes Lions*. Bethesda: Author. Retrieved from <http://news.marriott.com/2017/06/marriott-international-wins-two-creative-data-cannes-lions/>
- McKenna, R. (1995). *Real-Time Marketing*. Boston: Harvard Business School Publishing. Retrieved from <https://hbr.org/1995/07/real-time-marketing>

- McKenna, R. (1997). *Real Time: Preparing for the Age of Never Satisfied Customer*. Boston: Harvard Business School Press.
- McKinsey. (2016). *The CEO Guide to Customer Experience*. New York, NY: Author. Retrieved from <http://www.mckinsey.com/business-functions/operations/our-insights/the-ceo-guide-to-customer-experience>
- Mifsud, M., Cases, A., & N'Goala, G. (2015). Service appropriation: How do customers make the service their own? *Journal of Service Management*, 26(5), 706–725.
- Miles, M. B., Huberman, A. M., & Saldana, J. (2014). Qualitative data analysis: a methods sourcebook. In 3rd. Thousand Oaks, CA: SAGE.
- Minghetti, V., & Buhalis, D. (2010). Digital Divide and Tourism: Bridging the gap between markets and destinations. *Journal of Travel Research*, XX(X), 1–15.
- Mistilis, N., Buhalis, D., & Gretzel, U. (2014). 'eDestination Marketing of the future: The perspective of an Australian Tourism Stakeholder Network'. *Journal Travel Research*, 53, 1–13.
- Mobile Marketing Association (MMA). (2014). *Red Roof Inn Turns Flight Cancellations Into Customers*. New York, NY: MMA. Retrieved from http://www.mmaglobal.com/case-study-hub/case_studies/view/31739
- Neuhofer, B., Buhalis, D., & Ladkin, A. (2012). Conceptualising Technology Enhanced Destination Experiences. *Journal of Destination Marketing and Management*, 1(1), 36–46.
- Neuhofer, B., Buhalis, D., & Ladkin, A. (2013). A Typology of Technology-Enhanced Tourism Experiences. *International Journal of Tourism Research*, 16, 340–350.
- Neuhofer, B., Buhalis, D., & Ladkin, A. (2015). Smart technologies for personalised experiences. A case from the Hospitality Industry. *Electronic Markets*, 25(3), 243–254.
- Oliver, R., Roland, R., & Sajeev, V. (1998). Real-Time Marketing. *Marketing Management*, 7(4), 28–37.
- Paraskevas, A., Katsogridakis, I., Law, R., & Buhalis, D. (2011). Search Engine Marketing: Transforming Search Engines to Hotel Distribution Channels. *The Cornell Hotel Restaurant Administration Quarterly*, 52(2), 200–208.
- Patel, N. (2014). *The Psychology of Instant Gratification and How it Will Revolutionize your Marketing Approach* Irvine: Entrepreneur Media Inc. Retrieved from <https://www.entrepreneur.com/article/235088>
- Pine, J. B., & Gilmore, J. H. (1999). *The Experience Economy: Work is a Theatre and Every Business a Stage*. Cambridge: Harvard Business School.
- Porter, M. (1989). *Competitive Advantage. Creating and Sustaining Superior Performance*. London: The Free Press.
- Porter, M. E., & Heppelmann, J. E. (2014). How smart, connected products are transforming competition. *Harvard business review*, 92(11), 64–88.
- Prahalad, C. K., & Ramaswamy, V. (2004a). Co-Creation Experiences: The Next Practice in Value Creation. *Journal of Interactive Marketing*, 18(3), 5–14.
- Prahalad, C. K., & Ramaswamy, V. (2004b). Co-creating Unique Value with Customers. *Strategy and Leadership*, 32(3), 4–9.
- Pretzel Crisps. (2017). *We're Snack Geniuses*. Hanover: Author. Retrieved from <http://pretzelcrisps.com/our-story/>
- Reed, B., 2011, Disasters and emergencies: definitions, impacts and response. WEDC, Loughborough University. Retrieved from <https://wedc-knowledge.lboro.ac.uk/resources/booklets/G002-Disasters-and-emergencies-online.pdf>
- Reid, C. (2014). Real-Time Marketing Can Keep Business Successful – But How? *Computer/Information Science*, 37 (5), 6–10.
- Rihova, I., Buhalis, D., Gouthro, M., & Moital, M. (2018). Customer-to-customer co-creation practices in tourism: Lessons from Customer-Dominant logic. *Tourism Management*, 67, 362–375.
- Rihova, I., Buhalis, D., Moital, M., & Gouthro, M.-B. (2015). Conceptualising customer-to-customer co-creation in socially dense tourism contexts. *International Journal of Tourism Research*, 17(4), 356–363.
- Rust, R., & Oliver, R. (2000). The Real-time Service Product: Conquering Customer Time and Space. In J. Fitzsimmons & M. Fitzsimmons (Eds.), *New Service Development: Creating Memorable Experiences* (pp. 67–89). London: Sage.
- Saldana, J. (2016). *The Coding Manual for Qualitative Researchers*. 3E. London: Sage.
- Sarmah, B., Kamboj, S., & Kandampully, J. (2018). Social media and co-creative service innovation: An empirical study. *Online Information Review*, 42(7), 1146–1179.
- Sashi, C. (2012). Customer Engagement, Buyer-Seller Relationships, and Social Media. *Management Decision*, 50 (2), 253–272.
- Sawhney, M., Verona, G., & Prandelli, E. (2005). Collaborating to Create: The Internet as a Platform for Customer Engagement in Product Innovation. *Journal of Interactive Marketing*, 19(4), 4–17.
- Schade, M., Piehler, R., Warwitz, C., & Burmann, C. (2018). Increasing consumers' intention to use location-based advertising. *Journal of Product & Brand Management*, 27(6), 661–669.
- Schniederjans, D., Atlas, S., & Starkey, C. (2018). Impression management for corporate brands over mobile media. *Journal of Product & Brand Management*, 27(4), 385–403.
- Scott, D. M. (2012). *Real-time Marketing & PR: How to instantly engage your market, connect with customers, and create products that grow your business now*. Hoboken: John Wiley & Sons.
- Sigala, M., & Chalkiti, K. (2015). Knowledge management, social media and employee creativity. *International Journal of Hospitality Management*, 45, 44–58.
- Sigala, M., Christou, E., & Gretzel, U. (2012). *Social Media in Travel, Tourism and Hospitality: Theory, Practices and Cases*. Surrey: Ashgate Publishing Limited.
- Sinarta, Y., & Buhalis, D. (2018). Technology Empowered Real-Time Service. In B. Stangl & J. Pesonen (Eds.), *Information and Communication Technologies in Tourism* (pp. 283–295). Cham: Springer.
- Smart Focus ca. (2017). *What Does Real-Time Marketing Really Mean?* London: Smart Focus. Retrieved from <http://www.smartfocus.com/en/blog/what-does-real-time-marketing-really-mean>
- Smart Insights. (2014). *Real-time Marketing*. Leeds: Author. Retrieved from <http://www.smartinsights.com/online-pr/real-time-marketing-online-pr/>
- Spencer, A. J., Buhalis, D., & Moital, M. (2012). A hierarchical model of technology adoption for small owner-managed travel firms: An organizational decision-making and leadership perspective. *Tourism management*, 33(5), 1195–1208.
- Sullivan, L. (2011). *Pretzel Crisps Racks Up 4.2 Million Earned Media Impressions*. New York, NY: Media Post. Retrieved from <https://www.mediapost.com/publications/article/150409/pretzel-crisps-racks-up-42-million-earned-media-i.html>

- Sun, S., Fong, L. H. N., Law, R., & Luk, C. (2016). An Investigation of Gen-Y's Online Hotel Information Search: The Case of Hong Kong. *Asia Pacific Journal of Tourism Research*, 21(4), 443–456.
- Tiago, F., Couto, J., Faria, S., & Borges-Tiago, T. (2018). Cruise tourism: Social media content and network structures. *Tourism Review*, 73(4), 433–447.
- TranslateMedia. (2015). *Real-Time Marketing and It's Application in B2B Commerce*. London: Author. Retrieved from <https://www.translatemedia.com/translation-blog/real-time-marketing-and-its-application-in-b2b-commerce/>
- Tuškej, U., & Podnar, K. (2018). Consumers' identification with corporate brands: Brand prestige, anthropomorphism and engagement in social media. *Journal of Product & Brand Management*, 27(1), 3–17.
- Urde, M. (2016). The brand core and its management over time. *Journal of Product & Brand Management*, 25(1), 26–42.
- Vargo, S. L., & Lusch, R. F. (2004). Evolving to a New Dominant Logic for Marketing. *Journal of Marketing*, 68(1), 1–17.
- Veloutsou, C., & Guzmán, F. (2017). The evolution of brand management thinking over the last 25 years as recorded in the Journal of Product and Brand Management. *Journal of Product & Brand Management*, 26(1), 2–12.
- Viglia, G., Minazzi, R., & Buhalis, D. (2016). The influence of e-word-of-mouth on hotel occupancy rate. *International Journal of Contemporary Hospitality Management*, 28(9), 2035–2051.
- Visit Philadelphia. (2016). *VISIT PHILADELPHIA Wins Major Industry Award for Social Media*. Philadelphia: Author. Retrieved from <http://press.visitphilly.com/releases/visit-philadelphia-wins-major-industry-award-for-social-media>
- Wang, W., Wu, W., Luo, J., & Lu, J. (2017). Information technology usage, motivation, and intention: A case of Chinese urban senior outbound travelers in the Yangtze River Delta region. *Asia Pacific Journal of Tourism Research*, 22(1), 99–115.
- Wilson, A., Zeithaml, V. A., Bitner, M. J., & Gremler, D. D. (2012). *Services marketing: Integrating customer focus across the firm* (No. 2nd Eu). McGraw Hill.
- Xiang, Z., & Gretzel, U. (2010). Role of social media in online travel information search. *Tourism management*, 31(2), 179–188.
- Yelp. (2010). *I'm a Business Owner: What's a Yelp Check-in Offer?* San Francisco: Author. Retrieved from <https://www.yelpblog.com/2010/11/just-in-time-for-black-friday-check-in-offers>
- Yin, R. (2014). *Case Study Research: Design and Methods* (5th ed.). Thousand Oaks: Sage.
- Yoo, K.-H., & Gretzel, U. (2010, June 20–22). *Web 2.0: New Rules for Tourism Marketing*. Travel and Tourism Research Association 41st Annual Conference Proceedings 2010, San Antonio, Lake Orion, mi: Travel and Tourism Research Association. Retrieved from <http://ttra.omnibooksonline.com/2010/46322-ttra-1.5561/2010-t-001-1.5703/f-007-1.5743/26-a-1.5750>
- Yu Leung, X., Sun, J., & Bai, B. (2019). Thematic framework of social media research: State of the art. *Tourism Review*. doi:10.1108/TR-05-2018-0058
- Yuan, Y., Tseng, Y., & Ho, C. (2019). Tourism information technology research trends: 1990–2016. *Tourism Review*, 74, 5–19.
- Zhang, T., Bilgihan, A., Kandampully, J., & Lu, C. (2018). Building stronger hospitality brands through online communities. *Journal of Hospitality and Tourism Technology*, 9(2), 158–171.